Green Delivery Analytics: Advancements in Open Data for Sustainable Last Mile Logistics

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Abstract: In last mile logistics the implementation of electric vehicles and associated infrastructure are key elements to increase sustainability and efficiency. The Green Delivery Analytics project focuses on the systematic use of open data sources and compliance with open standards in data formats, structures and communication protocols to emphasize interoperability and adaptability. These standards facilitate seamless data access, exchange and publication, particularly to strengthen reusability, for example as contributions to national open data portals, such as the Mobilithek. This paper showcases how the project incorporates open data and associated interoperability architecture aspects to achieve more sustainability within the analytics and planning of last mile logistics in German cities. The aim is to advance the knowledge base on open data and interoperability in the field of last mile logistics and contribute to the UN SDG Targets 9.4, 11.6 and 13.2.

Keywords: Open Data, Last Mile Logistics, Data Architecture, Spatial Analytics, Smart City, Sustainable Logistics

1 Introduction

The 'last mile' in distribution logistics refers to the final segment of the supply chain, which is characterized by a high degree of complexity, partly due to the delivery of goods to a diverse range of recipients [SLWW13]. This distribution stage accounts for a significant portion of both the overall distribution costs and the associated emissions [Brab20]. The optimization of last mile logistics (LML) for electric vehicles, such as

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electric cargo bikes, is considered a central solution, as they contribute to both emission reduction and compliance with legal requirements [GBKP19]. The utilization of cargo bikes requires micro-hub structures for commissioning and consolidation of goods within cities [Trip19], as well as optimization of networks in regard to location, transportation, and route planning [HuHG20]. The economic and ecological implementation of cargo bikes and associated infrastructure requires a strategic combination of operational parameters [SBGM19] as well as the linking and analysis of patterns within large amounts of data from the contexts of city, sustainability and logistics [CBVV17].

The research project “Green Delivery Analytics” (GDA) takes on these challenges. Within the project a software tool will be developed, that enables municipalities and logistics service providers to plan and optimize the LML for the 80 biggest cities, based on a data-driven approach. [Bund24]

Comprehensive data focus forms the basis for data-driven decisions in logistics [GuMR21] and therefore the development of the data architecture is considered essential for the effective integration and processing of diverse data sources [Shei13] which can be used to realize the planning of sustainable LML. The high relevance and the large amounts of data in such software solutions require a particularly large focus on data architecture aspects such as data model, structure, format and management [Shei13].

While (open) data on logistics is rarely available, context data on e.g. demographics and city infrastructure is often more likely to be openly available based on data collection by municipalities [CVMT24]. This offers opportunities to address the challenges of data unavailability [AlSF25]. The usage and redistribution of open data and derived data products strives for the aforementioned focus on architectural aspects to fulfill the requirement of interoperability of open data and in this manner contribute to the implementation and monitoring of SDGs [Stat19].

RQ. How can an open data architecture be designed and implemented to integrate heterogeneous municipal, traffic and socio-economic datasets so that low-emission LML networks can be planned and evaluated for German cities?

This paper contributes to research on German LML through the development of an open data catalogue, a data architecture prototype and by enabling research on the implementation of UN SDG Targets 9.4, 11.6 and 13.2.

2 Background and Related Work

Due to the continuous growth in e-commerce activity and home deliveries, LML is playing an increasingly important role in freight logistics, especially in urban areas. Within the courier, express and parcel market, LML is the most expensive and least efficient segment of the logistics chain and accounts for a large proportion of emissions. [SLWW13]

The decisive influencing factors here are the logistics infrastructure, the consignment structure and the structure of the recipients. The EU is increasing pressure through climate policies and urban regulations to promote alternatives in LML [Euro00]. The layout of the delivery districts, the stop density and the proportion of standard, courier and express consignments also have an impact on efficiency and costs. Sustainable LML systems rely on the use of light commercial electric vehicles, cargo bikes, micro-hubs and urban consolidation centers, which require new approaches to infrastructure planning. [Keco25]

Network design for micro-vehicles must account for constraints such as limited range, lower payload, traffic regulations and topography. Traditional heuristics or planning models based on diesel vans are inadequate for LML [FeEs25]. Emerging tools support routing, fleet composition and hub placement using GIS and optimization models but often rely on proprietary software or siloed data [Elen23]. As [NeCA20] emphasizes, “the growing volume and variety of data produced in the urban ecosystem are crucial for obtaining the city's insights and building knowledge-based solutions for a smarter and more sustainable urban development”. The authors also state that open data can be considered an important enabler for such data-driven solutions [NeCA20]. The necessary data for LML solutions includes data on socio-demographics, geography as well as logistics/traffic data [GuMR21]. Some of these can be retrieved as open data.

Open data can be defined as data that can be freely used, reused and redistributed by anyone, typically under conditions that at most require attribution and adherence to share-alike licenses. The key principles of open data emphasize availability, accessibility, interoperability and clarity in licensing [NeCA20, Okf00a]. The Open Knowledge Foundation’s (OKF) definition on open data, adopted by related literature such as [NeCA20], underlines key attributes including universal participation and the freedom to access and redistribute data. The OKF’s full “Open Definition” extends this definition onto other domains, such as open knowledge or open formats [Okf00a, Okf00b]. In the context of geospatial standardization, the role of the Open Geospatial Consortium (OGC), as the authoritative body for open geospatial standards, is especially significant [Ogc00].

To ensure legal compliance with these principles, several licenses are recommended by the OKF, including Creative Commons Zero (CC0-1.0), Creative Commons Attribution (CC-BY-4.0), Open Data Commons Attribution (ODC-By-1.0) and the Open Database License (ODbL-1.0), among others. These licenses are evaluated for compatibility with the OKF’s principles in the Open Definition. [Okf00c]

The importance of open data is further supported by the FAIR Guiding Principles for scientific data management and stewardship, which aim to ensure that data should be FAIR (“Findable, Accessible, Interoperable and Reusable”). [WDAA16]

3 Catalogue of Open Data for LML

As shown in Chapter 2, LML optimization requires data from different domains, including socio-demographic, geographical, logistical/traffic, as well as geospatial reference data for statistical analytics [Bund00a]. Unfortunately, logistics and traffic data are usually not published as open data, due to economic and data protection reasons [TmEA19].

Table 1: Catalogue of Open Data for LML in Germany

Domain	Example Dataset	License	Source
Socio-demographic data	Official Census, Global Human Settlement Layer	dl-by-de/2.0, CC-By-4.0	[PSPF24, Stat00]
Geographical data	Official Federal Data (ATKIS, ALKIS), OpenStreetMap	dl-zero-de/2.0 or dl-by-de/2.0, CC-By-4.0, ODbL-1.0	[Advd00, Lgln00, MoMi17]
Logistical data	NA	NA	NA
Geospatial Reference data	GeoGitter INSPIRE (ETRS89-extended LAEA)	dl-by-de/2.0	[Bund00a]

As seen in Table 1, open data can be derived from various sources, ranging from official federal cadastral offices or official European projects to volunteered geographic information in OpenStreetMap. Problems arise through the complexity in used licenses (even within datasets, due to state-level organization), which may require name attribution or even share-alike.

4 Prototype Architecture

Due to the project's objective to publish its own data to the German open data portal for mobility data – Mobilithek [Bund00b] – it is needed to design the projects data architecture in a way that ensures reusability of produced data [WDAA16]. The FAIR Guiding Principles showcase why data architecture plays a big role in the FAIRness of publishable data. Subgoals of the FAIR Guiding Principles include ensuring that the used "protocol is open, free and universally implementable", data is described in "formal (…) and broadly applicable language" or that data meets "domain-relevant community standards" [WDAA16]. In general, data architecture can be understood as the blueprint for how all data is collected, stored, integrated and made accessible for analysis [Shei13]. To ensure the project's data architecture is designed accordingly, used file formats, data structures and communication protocols need to be coherent and comprehensible. A well-designed architecture can not only establish the FAIR Guiding Principles but also enables a seamless integration of the different datasets that vary greatly in structure and granularity.

Figure 1 gives a brief overview of the data architecture used within the project GDA to integrate open data and contribute to interoperability of the project's approach. It shows the architecture divided into different data processing layers. Within the data storage layer an object-relational database including a spatial extension is used as a central and harmonized data store of the logistical and contextual data.

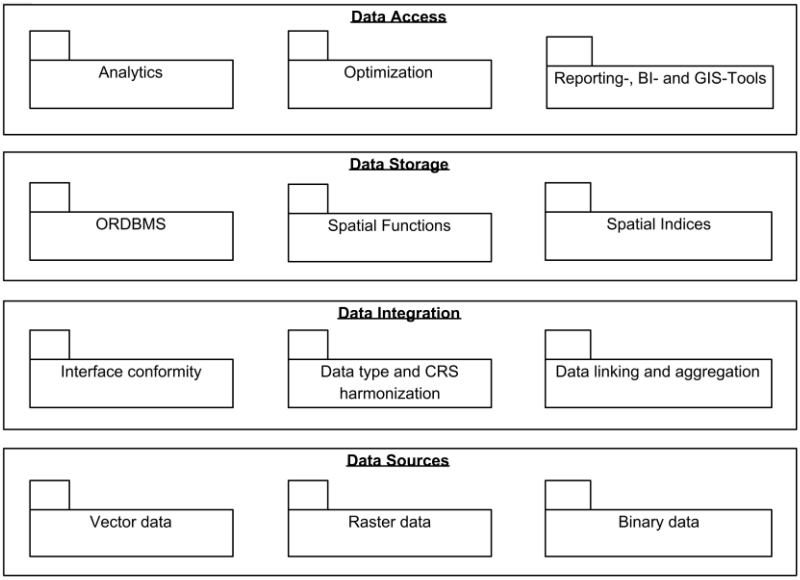

Fig. 1: Overview of the Prototype Architecture of the GDA Project

Two data models exist for the storage of spatial information, vector and raster. Vector data consists of points, which might form polygons or lines, and is useful for the rendering of data, as the shape of an object can be accurately displayed. Raster or grid data is stored in geolocated grid cells or pixels. The examples can range from high-resolution satellite imagery to population densities in square kilometers. Due to the rasterization of data some form of aggregation usually takes place, while the vector data, which explicitly stores geometries for individual objects, usually needs to simplify the number of points used to shape a polygon or line. [CoKa92]

Both data models utilize different data formats, with formats describing "a specific, pre-established structure for the organization of a digital file or bitstream" [Prem15]. The formatting of information has been a focus of standardization processes, as big data analytics and data exchange requires consistent and unique conventions [GCLM17].

In the current draft of the Open Knowledge Foundation's Open Format Definition an open format is defined as "a freely available published specification which places no restrictions, monetary or otherwise, upon its use" or at the very least as a format that "can be processed with at least one free/libre/open-source software tool" [Okf00d]. The GDA architecture uses open file formats wherever possible to avoid vendor lock-in effects and enhance interoperability.

The Project uses the following open formats [Gisg19]:

1. Raster, such as GeoTIFF.
2. Vector, such as Shapefile, GeoPackage, GML, GeoJSON.
3. Binary, such as OSM PBF.

In this project, as shown in the previous sections, raster data and vector data have to be handled simultaneously to enable a complex analysis of geospatial and socio-demographic data. Therefore, the project must be able to harmonize and unify the heterogeneous formats in terms of resolution, attribution and reference systems for example through the rasterization of vector data or the vectorization of raster data. [SPLL20]

The harmonization of coordinate reference systems (CRS) plays a crucial role in the integration of geospatial datasets, as these datasets are often based on heterogeneous reference systems. A CRS provides a standardized framework to describe locations on Earth's surface by using coordinate definitions. This may include geographic (e.g., latitude/longitude), projected (planar), or geocentric (3D) coordinate systems. [KSSP23]

Since many datasets are published using varying CRS, inconsistencies may arise in spatial analyses if no harmonization is conducted. Particularly for applications involving distance, area or direction calculations, projected CRS with minimal regional distortion are preferred. In this context, the use of EPSG:3035 (ETRS89 / LAEA Europe) was chosen due to its suitability for pan-European statistical and area analyses, as well as its compatibility with INSPIRE-recommended standards, in contrast to the official geodetic CRS EPSG:4258, which is not projected and less suitable for metric operations. [ALGI01]

To enable necessary spatial transformations a flexible and robust data management system is essential. The project utilizes PostgreSQL, an open source object-relational database system, to store and process analytical datasets, which include fact tables (e.g. stop density or cost metrics), dimension tables (e.g. road types, surface characteristics) and shared spatial and temporal dimensions (e.g. city grid cells, time intervals). PostgreSQL uses a "BSD-style license and is thus free and open source software" [AgRa16] and has been one of the first databases to allow for spatial analysis. [AgRa16]

PostGIS, one of the many available extensions for PostgreSQL, offers specialization for spatial analysis and supports spatial joins (e.g. intersect, within, distance-based joins), spatial indexing (e.g. GiST indexes) and operations such as buffering, clipping and coordinate transformations. [AgRa16]

Besides efficient data storage and processing, the interoperability and accessibility of spatial datasets also depend significantly on the utilization of standardized and open protocols. In general, open protocols are a specific type of open standard, just like open formats, and should ensure the interoperability between systems, services and stakeholders

[CeFu07]. Protocols are central to the FAIR Guiding Principle of Accessibility, which states that (meta)data should be retrievable through standardized communication protocols that are open, free, and universally implementable, while also supporting authentication and authorization mechanisms. [WDAA16]

In the project, protocols are necessary for accessing the data sources, communication between internal services or components, as well as publishing results to the Mobilithek. Even though many open protocols for spatial formats have been developed, many services still rely on HTTP Downloads (like ATKIS, Census, Global Human Settlement Layer) or OGC Web Feature Service or Web Map Service (like ALKIS). For futureproof interoperability, tools should be able to use OGC API (currently establishing), REST APIs and GraphQL.

5 Conclusion

The study has shown that a robust open data architecture for sustainable last-mile logistics has to address a variety of standards to be able to systematically integrate spatial structure layers, socio-demographic data and analytical grids into a PostGIS-centered stack that complies with FAIR principles. The GDA project delivers a systematic overview of open data sources in the context of LML analytics. Furthermore, open formats, open protocols as well as open relational and spatial data structures that are associated with these data sources are described to briefly present the requirements of processing this data. The presented content is particularly useful for researchers, urban planners and logistics stakeholders who aim to develop or enhance data-driven solutions for sustainable LML. It also serves as a valuable foundation for software developers, data engineers and system architects who intend to process or analyze such data.

At the same time, the paper has highlighted several challenges, e.g. data sparsity on logistics-specific data and the heterogeneity of existing open standards. Nevertheless, the modular architecture and adherence to open protocols aim for flexibility and expandability. Next steps include the definition of specific data integration components, the ingesting of carrier-provided logistics data, the utilization of GraphQL/REST endpoints for logistics/traffic data (not open data) as well as deploying AI-enabled analytics that transform curated datasets into predictive demand and enable the optimization of location, transportation and route planning.

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